An extensibly causal quantum process with no purification

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The process-matrix formalism describes two laboratories connected in a way that does not presuppose which of them acts first. Two conditions have been proposed to single out the "physical" processes. One is observational: a process is extensibly causal if the statistics of every experiment, including experiments assisted by arbitrary shared entanglement, can be explained by a mixture of the two causal orders. The other is structural: a process is purifiable if it is the restriction of a larger process that carries reversible local operations to a reversible global evolution. Every purifiable bipartite process is extensibly causal, and an open problem exists as to whether the converse holds. We show that the converse is false, by exhibiting a process on four qubits that is extensibly causal but not purifiable.

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Results were formalized in Lean 4 by AI agents and checked by Lean's kernel; the formalization is at `https://github.com/physical-superintelligence/oqp-batch-2026/tree/main/oqp43-extensible-causality/lean`.

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In ordinary quantum mechanics, an experiment involving two laboratories comes with a causal structure: Alice acts before Bob, or Bob before Alice, or the two are spacelike separated and neither can signal to the other. Oreshkov, Costa, and Brukner [1] asked what the most general way to connect two laboratories is if one keeps the quantum formalism inside each laboratory but gives up a global causal order. The answer is a process matrix: a positive operator S that assigns a probability to every pair of local operations, generalizing both a bipartite state and a channel from one laboratory to the other. Surprisingly, some valid process matrices are not mixtures of the two causal orders, and some of these produce correlations violating "causal inequalities," the analog for causal order of Bell's inequalities [1, 2].

No physical realization of such correlations is known, which raises the question of what additional condition singles out the physical process matrices. Two conditions have been proposed, and the purpose of this Letter is to show that they are not equivalent. The first is a condition on the probabilities that experiments produce. A process is *causal* if the statistics of every experiment performed on it can be explained as a mixture of "Alice acts first" and "Bob acts first," and it is *extensibly causal* if this remains true when the two laboratories are supplied with an arbitrary shared ancilla, which may be entangled [3]. The second condition, proposed by Araújo, Feix, Navascués, and Brukner [4], is a condition on the process itself, modeled on two basic facts of ordinary quantum mechanics: every mixed state is the marginal of a pure state, and every channel is the restriction of a unitary evolution. A process is *purifiable* if it is the restriction of a larger process that is reversible, in the sense that it maps reversible local operations to a reversible global evolution. A third notion sits between these two. A process is *causally separable* if it is itself a mixture of an Alice-first process and a Bob-first process. Such a process is clearly extensibly causal. It is also purifiable, by purifying each ordered component and letting a control qubit choose between them; and conversely, a theorem of Refs. [5, 6] states that every purifiable bipartite process is causally separable. So for two laboratories the first two notions coincide, and the logical picture is that of Fig. 1. Problem 43 of the IQOQI list of open problems in quantum information asks whether the picture collapses further: whether every extensibly causal process is purifiable [7].

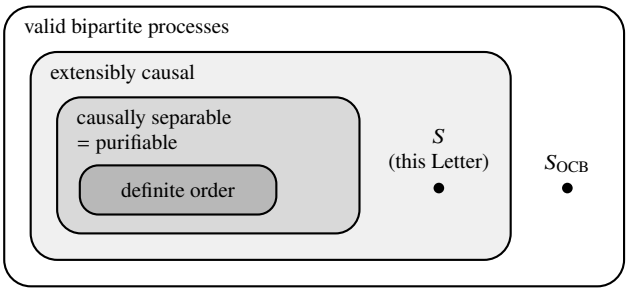


FIG. 1. Nesting of the three properties for a bipartite process with trivial global past and future. Purifiability and causal separability coincide and lie strictly inside extensible causality; the process S constructed here lies in the gap. The process S_{OCB} of Ref. [1] violates a causal inequality and lies outside all three.

We will show that the answer is no: there is a process on four qubits, which is extensibly causal but not purifiable. The idea is simple to state. For a causally separable process,

one decomposition of the operator explains all experiments at once, with the same weight λ in every experiment. Extensible causality asks for less. It requires only that the probability table of each experiment be some mixture of an Alice-first table and a Bob-first table; the two tables and the weight may be chosen afresh for each experiment, and they need not be the statistics of any process matrix. Our example lives in the gap between these two requirements. Feix, Araújo, and Brukner [8] showed that causally nonseparable processes exist whose statistics are nevertheless causal in every experiment without ancillas, and found numerical evidence that some of them remain causal with entangled ancillas, but were unable to prove it; the difficulty is to control arbitrary measurements on arbitrary ancillas at once. The main new ingredient here is an operator identity of sum-of-squares type that does precisely this.

REVIEW OF PROCESSES AND CAUSAL ORDER

Consider an experiment in which Alice chooses a setting x and records an outcome a, while Bob chooses y and records b. Whatever connects the laboratories, all that is observed is the conditional distribution $p(a, b|x, y)$. We call such a distribution *causal* if

$$p = \lambda\, p^{A\prec B} + (1-\lambda)\, p^{B\prec A}, \qquad 0 \le \lambda \le 1, \tag{1}$$

where $p^{A\prec B}$ and $p^{B\prec A}$ are conditional distributions such that Alice's marginal of $p^{A\prec B}$ does not depend on y, and Bob's marginal of $p^{B\prec A}$ does not depend on x. Here λ is a single number, the same for all settings. The condition on $p^{A\prec B}$ only forbids signaling from Bob to Alice; Bob's outcome may depend on Alice's setting. Causality is therefore a much weaker requirement than Bell locality, which demands that $p(a, b|x, y) = \sum_\mu q(\mu)\, p_A(a|x, \mu)\, p_B(b|y, \mu)$ for some classical variable μ shared by the laboratories, and so forbids signaling in both directions and more. A causal inequality is a linear inequality satisfied by all causal distributions [1, 2].

Now let the laboratories be quantum. Alice's has an input Hilbert space A_I and an output space A_O, and Bob's has B_I and B_O; all four are finite dimensional, and in our example they are qubits. The most general thing Alice can do is apply a quantum instrument: a family of completely positive maps $\mathcal{M}_{a|x}$ from A_I to A_O, labeled by her setting x and outcome a, whose sum over a is trace preserving. We represent $\mathcal{M}_{a|x}$ by its Choi matrix $M_{a|x}$, a positive operator on $A_I \otimes A_O$ normalized so that $\sum_a \mathrm{Tr}_{A_O} M_{a|x} = I_{A_I}$.[1] Bob's instrument is represented likewise by operators $N_{b|y}$ on $B_I \otimes B_O$. A process matrix is an operator S on $A_I \otimes A_O \otimes B_I \otimes B_O$ such that

$$p(a, b|x, y) = \mathrm{Tr}\big[S\,(M_{a|x} \otimes N_{b|y})\big] \tag{2}$$

is a probability distribution for every pair of instruments. Thus $S \ge 0$, and $\mathrm{Tr}[S(M \otimes N)] = 1$ whenever M and N are Choi matrices of quantum channels (completely positive and trace-preserving maps). The normalization condition amounts to linear constraints on S, written out in [1], which forbid any term by which a laboratory could influence its own input, or by which the two laboratories influence each other simultaneously; we will verify them directly for our example. A process S is *causal* if (2) is causal for all instruments, and *extensibly causal* [3] if, moreover, $S \otimes \sigma$ is causal for every density matrix σ on any pair of ancillary spaces $R_A \otimes R_B$, where now Alice's instruments act on $A_I \otimes R_A$ and Bob's on $B_I \otimes R_B$. Neither the ancilla dimensions nor the number of settings and outcomes are restricted, and the associated decomposition (1) may depend on σ and on the whole collection of instruments. In other words, extensible causality constrains S only through its outputs: every table $p(a, b|x, y)$ that S can produce must admit some decomposition of the form (1), and nothing is asked of the form of S itself.

Two familiar process matrices will be needed. If $S = \rho \otimes I_{A_O} \otimes I_{B_O}$ for a density matrix ρ on $A_I \otimes B_I$ (tensor factors are always understood in the order A_I, A_O, B_I, B_O), Alice and Bob measure a shared state and their outputs are discarded; there is no signaling. If instead $S = \widehat{S} \otimes I_{B_O}$ with $\widehat{S} \ge 0$ on $A_I \otimes A_O \otimes B_I$ and $\mathrm{Tr}_{B_I} \widehat{S} = \rho \otimes I_{A_O}$ for a density matrix ρ on A_I, then Alice receives the state ρ, her output is sent through a channel to Bob's input, and Bob's output is discarded. This is the general form of a process in which Alice acts before Bob [9]; we say that S has order $A \prec B$. A valid S has order $A \prec B$ precisely when it does not depend on B_O, which we write as $S = \mathcal{R}_{B_O} S$, where for any tensor factor Y the "replacement" map is $\mathcal{R}_Y(T) = \mathrm{Tr}_Y T \otimes I_Y/\dim Y$. Likewise S has order $B \prec A$ precisely when $S = \mathcal{R}_{A_O} S$. A process is *causally separable* if it is a convex combination

$$S = \lambda\, S^{A\prec B} + (1-\lambda)\, S^{B\prec A} \tag{3}$$

of processes of the two orders. Such a process is causal in every experiment, with or without ancillas, since the two terms of (3) produce the two terms of (1) with the same λ. Thus, every causally separable process is extensibly causal. The converse is what we show to be false in this letter.

Purification.—Suppose that, besides the four laboratory spaces, a process has a global past P and a global future F. The process is then a positive operator $\widetilde{S}$ on $P \otimes A_I \otimes A_O \otimes B_I \otimes B_O \otimes F$ with the property that inserting channels in the two laboratories produces a channel from P to F [4, 9]. Following [4], we say that $\widetilde{S}$ is *pure* if, whenever the two local operations are unitary, the resulting map from P to F is unitary. The local unitaries may act also on auxiliary systems passing through the laboratories, so purity concerns reversibility in all circumstances.

It is easy to show that a pure $\widetilde{S}$ also must be rank one. To see this, remember that the induced map must be unitary for every choice of local unitaries, and so we may test it on a choice. For example, let Alice apply the unitary that swaps her input out to an external system $E'_A \cong A_I$ and an external system $E_A \cong A_O$

[1] We use the convention of [1], in which M is the transpose of the usual Choi matrix $\sum_{ij} |i\rangle\langle j| \otimes \mathcal{M}(|i\rangle\langle j|)$; nothing below depends on this.

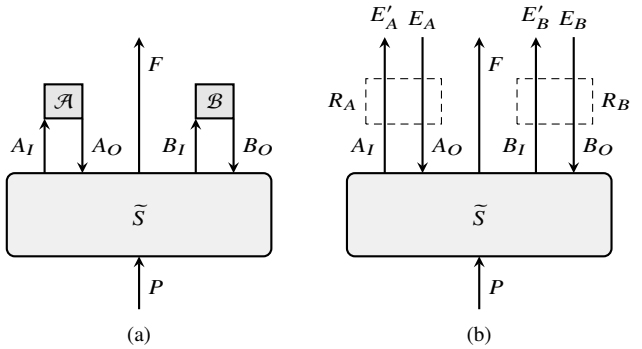


FIG. 2. (a) A process $\widetilde{S}$ with global past P, global future F, and two laboratory slots: Alice's operation $\mathcal{A}$ receives A_I and returns A_O, and likewise Bob's. (b) With the swaps R_A, R_B of Eq. (4) inserted (dashed), each input is routed straight out to E'_A, E'_B and the external systems E_A, E_B straight in. The map from $P\otimes E_A\otimes E_B$ to $F\otimes E'_A\otimes E'_B$ is $\widetilde{S}$ itself with its legs renamed, so purity makes $\widetilde{S}$ the Choi operator of a unitary, Eq. (5).

in to her output,

$$R_A\,|i\rangle_{A_I}|o\rangle_{E_A} = |o\rangle_{A_O}|i\rangle_{E'_A}, \tag{4}$$

and let Bob apply R_B likewise (Fig. 2). Here and below, for a linear map $M\colon X\to Y$ we write $|M\rangle\!\rangle = \sum_k |k\rangle_X \otimes M|k\rangle$ for its Choi vector, so that $|M\rangle\!\rangle\langle\!\langle M|$ is the Choi matrix of $\rho\mapsto M\rho M^\dagger$ and $|I\rangle\!\rangle_{XY} = \sum_k |k\rangle_X|k\rangle_Y$. The Choi vector of the swap is a pair of identity states, $|I\rangle\!\rangle_{A_I E'_A}|I\rangle\!\rangle_{E_A A_O}$, and contracting an index of $\widetilde{S}$ against one of them merely renames it. So the map induced from $P\otimes E_A\otimes E_B$ to $F\otimes E'_A\otimes E'_B$ has Choi operator $\widetilde{S}$ itself,

$$\widetilde{S} = |\mathcal{U}\rangle\!\rangle\langle\!\langle\mathcal{U}|, \quad \mathcal{U}\colon P\otimes A_O\otimes B_O \to A_I\otimes B_I\otimes F, \tag{5}$$

with $\mathcal{U}$ unitary. The converse fails: a rank-one $\widetilde{S}$ only carries unitary local operations to an isometry, which may enlarge the space, and if that were allowed every process would be purifiable [4]. A process S with trivial past and future is *purifiable* if it is obtained from a pure $\widetilde{S}$ by feeding a fixed state into the past and discarding the future, $S = \mathrm{Tr}_F[\langle 0|_P\,\widetilde{S}\,|0\rangle_P]$, with P and F of arbitrary finite dimension.

Processes with a definite causal order are purifiable, by the Stinespring dilation of the circuit that realizes them [9]. A mixture of processes as in (3) is also purifiable, by the mechanism of the quantum switch. By (5), write $\widetilde{S}_A = |s_A\rangle\!\rangle\langle\!\langle s_A|$ and $\widetilde{S}_B = |s_B\rangle\!\rangle\langle\!\langle s_B|$ for purifications of $S^{A\prec B}$ and $S^{B\prec A}$, with pasts P_A, P_B and futures F_A, F_B, and let $|v_A\rangle\!\rangle$ on $P_A\otimes F_A$ be the vector obtained from $|s_A\rangle\!\rangle$ by inserting identity operations in both laboratories; it represents a unitary $V_A\colon P_A\to F_A$. Define $|v_B\rangle\!\rangle$ likewise. Add a control qubit C to the past and C' to the future, and set

$$\begin{aligned}|s\rangle\!\rangle = {}& |\phi_0\rangle_C|0\rangle_{C'}\,|s_A\rangle\!\rangle|v_B\rangle\!\rangle \\ &+ |\phi_1\rangle_C|1\rangle_{C'}\,|v_A\rangle\!\rangle|s_B\rangle\!\rangle,\end{aligned} \tag{6}$$

with $|\phi_0\rangle = \sqrt{\lambda}|0\rangle + \sqrt{1-\lambda}|1\rangle$ and $|\phi_1\rangle\perp|\phi_0\rangle$. Inserting unitary local operations into $\widetilde{S} = |s\rangle\!\rangle\langle\!\langle s|$ gives the map

$$G = |0\rangle\langle\phi_0|\otimes G_A\otimes V_B + |1\rangle\langle\phi_1|\otimes V_A\otimes G_B, \tag{7}$$

where G_A, G_B are the unitaries induced by $\widetilde{S}_A, \widetilde{S}_B$. So G is unitary, and $\widetilde{S}$ is pure (it maps channels to channels by the same block structure). Feeding $|0\rangle$ into C selects the amplitudes $\langle 0|\phi_0\rangle = \sqrt{\lambda}$ and $\langle 0|\phi_1\rangle = \sqrt{1-\lambda}$, and tracing out F kills the cross terms, the two branches being orthogonal on C'; since $\langle 0|V_B^\dagger V_B|0\rangle = 1$, the result is $\mathrm{Tr}_F[\langle 0|_P S|0\rangle_P] = \lambda S^{A\prec B} + (1-\lambda)S^{B\prec A}$, as required. The converse is the theorem of Yokojima, Quintino, Soeda, and Murao [5] and of Barrett, Lorenz, and Oreshkov [6]: *a purifiable bipartite process with trivial past and future is causally separable.* Its proof starts from (5) and shows that the constraints forbidding each laboratory from influencing its own input split P into sectors, in each of which the order is definite; discarding F destroys the coherence between sectors, leaving a mixture (3).

Thus, for two laboratories, purifiability and causal separability are the same property, and it will suffice therefore to show that our extensibly causal process is not causally separable. A convenient tool for that is a causal witness [10]: a Hermitian operator W with $\mathrm{Tr}(WS') \geq 0$ for every causally separable S'. A sufficient condition is that $\mathcal{R}_{B_O}W \geq 0$ and $\mathcal{R}_{A_O}W \geq 0$, for if S' has order $A\prec B$ then $\mathrm{Tr}(WS') = \mathrm{Tr}(W\mathcal{R}_{B_O}S') = \mathrm{Tr}[(\mathcal{R}_{B_O}W)\,S'] \geq 0$, the map $\mathcal{R}_{B_O}$ being self-adjoint with respect to the trace pairing; similarly for the other order, and then for mixtures.

A NON-CAUSALLY SEPARABLE BUT EXTENSIBLY CAUSAL PROCESS

We now turn to explaining the counterexample in detail. We will find a causally separable witness W and an extensibly causal process with operator S such that

$$\mathrm{Tr}(SW) \leq 0. \tag{8}$$

Let I, X, Z be the identity and Pauli matrices on a qubit, and let a string of four Paulis denote the tensor product in the order $A_I A_O B_I B_O$ (i.e. $X\otimes X\otimes I\otimes Z$ is denoted by $XXIZ$). Our process is

$$\begin{aligned}S = \frac{1}{80}\big(&20\,IIII + 10\,ZIII + 10\,IIZI \\ &+ 5\,ZIZI + 4\,IZZI + 4\,ZIXZ\big).\end{aligned} \tag{9}$$

Compare the process $S_{\mathrm{OCB}} = \frac{1}{4}\big[IIII + (IZZI + ZIXZ)/\sqrt{2}\big]$ of [1], which violates a causal inequality. The last two terms of (9) are the signaling terms of S_{OCB}: $IZZI$ sends Alice's output, in the Z basis, to Bob's input, and $ZIXZ$ sends Bob's output to Alice's input, conditioned on the X component of Bob's input. Here they carry the smaller coefficient $1/20$, and the first four terms, which add up to $\theta\otimes I\otimes\theta\otimes I$ with $\theta = \frac{1}{4}(3|0\rangle\langle 0| + |1\rangle\langle 1|)$, describe a biased rather than maximally mixed state on the inputs. The bias is essential to the argument below. To our

knowledge S has not been considered before. The nonseparable processes with causal models of Ref. [8] are built instead from a depolarizing channel and $ZIXZ$, and their causal models rest on a partial-transposition argument that neither extends to entangled ancillas nor applies to S, which is invariant under partial transposition of Bob's systems.

Theorem. *The operator S of* (9) *is a valid process matrix of full rank. It is extensibly causal, but is not causally separable. Consequently it has no purification, whatever the dimensions of P and F.*

First we check that S is a valid process matrix. Let M and N be Choi matrices of channels, so that $\mathrm{Tr}_{A_O} M = I$ and $\mathrm{Tr}_{B_O} N = I$. The term $IZZI$ contributes $\mathrm{Tr}[(IZ)_{A_I\otimes A_O} M]\ \mathrm{Tr}[(ZI)_{B_I\otimes B_O} N]$, and the second factor is $\mathrm{Tr}[Z\ \mathrm{Tr}_{B_O} N] = \mathrm{Tr}\, Z = 0$; the term $ZIXZ$ contributes zero for the same reason, with the roles of the laboratories exchanged. The first four terms contribute $\mathrm{Tr}[\theta\,\mathrm{Tr}_{A_O} M]\ \mathrm{Tr}[\theta\,\mathrm{Tr}_{B_O} N] = 1$. The same computation applies to $S\otimes\sigma$ in the presence of ancillas. For positivity, every term in (9) is diagonal in the computational basis except the X in the last one, so S is a direct sum of 2×2 blocks on B_I. Up to the sign of the off-diagonal entry, the blocks of $80S$ are

$$\begin{pmatrix} 49 & 4 \\ 4 & 11 \end{pmatrix},\ \begin{pmatrix} 41 & 4 \\ 4 & 19 \end{pmatrix},\ \begin{pmatrix} 19 & 4 \\ 4 & 1 \end{pmatrix},\ \begin{pmatrix} 11 & 4 \\ 4 & 9 \end{pmatrix}, \tag{10}$$

with determinants 523, 763, 3, and 83. So $S > 0$, with smallest eigenvalue $(10-\sqrt{97})/80 \approx 0.0019$.

S is not causally separable

A simple calculation shows that S is not causally separable. Let $P_1 = |1\rangle\langle 1|$ and $D = I + 8P_1 = \mathrm{diag}(1, 9)$, and set

$$W = P_1\, I\, D\, I - P_1\, Z\, (DZ)\, I + 3\, P_1\, I\, X\, Z. \tag{11}$$

We claim that W is a causal witness. Applying $\mathcal{R}_{B_O}$ kills the last term and leaves $P_1\otimes(I\otimes D - Z\otimes DZ)\otimes I$. On the two eigenspaces of Z on A_O, the operator on B_I is $D(I\mp Z)$, that is, $18P_1$ or $2|0\rangle\langle 0|$; both are positive. Applying $\mathcal{R}_{A_O}$ instead kills the middle term and leaves $P_1\otimes I\otimes(D\otimes I + 3X\otimes Z)$. On the two eigenspaces of Z on B_O, the operator on B_I is $D\pm 3X = \left(\begin{smallmatrix} 1 & \pm 3 \\ \pm 3 & 9 \end{smallmatrix}\right)$, which is positive and of rank one. On the other hand, writing $P_1 = (I-Z)/2$ and $D = 5I - 4Z$ and using the orthogonality of distinct Pauli strings, one finds that the three terms of (11) contribute 3, -2, and $-6/5$ to $\mathrm{Tr}(WS)$, so that

$$\mathrm{Tr}(WS) = -\tfrac{1}{5} < 0. \tag{12}$$

Thus S is not causally separable, and by the theorem of [5, 6] it is not purifiable. Note that (12) concerns the operator S; it does not say that S violates a causal inequality, and we now show that it violates none, even with entangled ancillas.

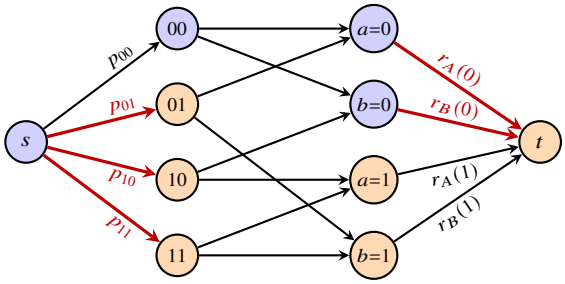


FIG. 3. The network behind Eq. (14), drawn for two outcomes per party. Fluid enters at the source s, reaches the outcome nodes through pipes of capacity p_{ab}, flows on through pipes of unlimited capacity to the row node a and the column node b of each outcome, and leaves through pipes of capacity $r_A(a|x)$, $r_B(b|y)$ into the sink t. A division (14) is a way of sending one unit through this network. The colors show the cut with $\alpha = \beta = \{0\}$: blue nodes (s, the outcome 00, and the row and column nodes in α and β) are those still connected to the source after the cut, orange nodes are those cut off. The removed pipes, in red, are exactly the ones running from a blue node to an orange node, $s\to 01$, $s\to 10$, $s\to 11$, $a{=}0\to t$ and $b{=}0\to t$, of total capacity $r_A(0|x) + r_B(0|y) + 1 - p_{00}$, which Eq. (13) makes at least one. Pipes from orange to blue nodes are not removed.

S is extensibly causal

Now we demonstrate that S is in fact extensibly causal. The argument has two steps: a general criterion, Lemma 1, which guarantees extensible causality of the process, and the verification, Lemma 2, that S meets this criterion in every experiment.

Lemma 1. *Let S be a bipartite process and $\lambda\in(0,1)$. Suppose that for every ancilla state σ and every pair of instruments there are nonnegative numbers $r_A(a|x)$ and $r_B(b|y)$, the first depending only on x and the second only on y, with $\sum_a r_A(a|x) = \lambda$ and $\sum_b r_B(b|y) = 1-\lambda$, such that for all sets α of Alice's outcomes and β of Bob's outcomes*

$$p(\alpha,\beta|x,y) \le r_A(\alpha|x) + r_B(\beta|y), \tag{13}$$

with sums over $a\in\alpha$ and $b\in\beta$ being understood. Then S is extensibly causal.

Proof. Fix an experiment and settings x, y, and write p_{ab} for $p(a,b|x,y)$. We want to divide the probability p_{ab} of each outcome (a,b) into a part z_{ab} credited to Alice's outcome a and a remainder $p_{ab} - z_{ab}$ credited to Bob's outcome b, such that the total credited to each a is $r_A(a|x)$ and the total credited to each b is $r_B(b|y)$:

$$0 \le z_{ab} \le p_{ab},$$
$$\sum_b z_{ab} = r_A(a|x), \qquad \sum_a (p_{ab} - z_{ab}) = r_B(b|y). \tag{14}$$

The totals are consistent, since $\sum_{ab} p_{ab} = 1 = \sum_a r_A(a|x) + \sum_b r_B(b|y)$. Whether such a division exists can be recast as a question about pipes (Fig. 3). Picture a source connected to one

node per outcome (a, b) by a pipe of capacity p_{ab}. Connect each outcome node by pipes of unlimited capacity to a node for its row a and a node for its column b (as illustrated in Fig. 3). Furthermore, connect each row node a and column node b to a sink by pipes of capacity $r_A(a|x)$ and $r_B(b|y)$.

A *flow* assigns to each pipe an amount of fluid between zero and its capacity, such that at every node other than the source and the sink the fluid entering equals the fluid leaving; the *value* of the flow is the total amount leaving the source node. It is not hard to see that a flow of value one is the same thing as a division as in (14). The source pipes have total capacity $\sum_{ab} p_{ab} = 1$, so a flow of value one fills each of them exactly and delivers p_{ab} to the node (a, b). Conservation at that node means it passes some amount $z_{ab} \in [0, p_{ab}]$ on to its row node and the remainder $p_{ab} - z_{ab}$ to its column node. Conservation at the row node a means that the pipe from a to the sink carries $\sum_b z_{ab}$, which cannot exceed its capacity $r_A(a|x)$; likewise $\sum_a (p_{ab} - z_{ab}) \le r_B(b|y)$. Finally, the sink pipes must carry one unit in total while their capacities add up to $\sum_a r_A(a|x) + \sum_b r_B(b|y) = 1$, so in fact these inequalities are all equalities. In other words, z_{ab} must obey the equations in (14).

To show that the network can support a flow of value one, we use the max-flow min-cut theorem together with the inequality (13). A *cut* is a division of the nodes into a source side containing s and a sink side containing t (blue and orange in Fig. 3), and its capacity is the total capacity of the pipes leading from the source side to the sink side; pipes in the other direction do not count. Every unit of fluid that reaches t must cross from the source side to the sink side, so no flow has value larger than the capacity of any cut. The max-flow min-cut theorem [11] says that the largest flow value through the network is exactly the smallest cut capacity. Hence a flow of value one exists if and only if every cut has capacity at least one, and it remains to list the cuts. Since the intermediate pipes have infinite capacity, a cut of finite capacity never has an outcome node on the source side while its row node or column node is on the sink side. Let α be the set of rows and β the set of columns on the source side. An outcome (a, b) with $a \notin \alpha$ or $b \notin \beta$, that is, with $(a, b) \notin \alpha \times \beta$, must then be on the sink side, so its source pipe crosses the cut; and the sink pipes of the rows in α and the columns in β cross it too. The capacity is therefore at least

$$r_A(\alpha|x) + r_B(\beta|y) + \big[1 - p(\alpha, \beta|x, y)\big], \tag{15}$$

the bracket being the total capacity of the source pipes of the outcomes outside $\alpha \times \beta$, with equality when every outcome of $\alpha \times \beta$ is on the source side, as in Fig. 3. That (15) is always at least one is exactly the inequality (13); the value one is attained by the cut with only s on the source side ($\alpha = \beta = \emptyset$), whose capacity is $\sum_{ab} p_{ab} = 1$. Therefore the minimum cut capacity is one, a flow of value one exists, and the division (14) follows.

Given this division z_{ab}, set $p^{A\prec B} = z/\lambda$ and $p^{B\prec A} = (p - z)/(1 - \lambda)$. These are normalized distributions with $p = \lambda\, p^{A\prec B} + (1 - \lambda)\, p^{B\prec A}$; Alice's marginal of $p^{A\prec B}$ is $r_A(a|x)/\lambda$, independent of y, and Bob's marginal of $p^{B\prec A}$ is $r_B(b|y)/(1 - \lambda)$, independent of x. The weight λ and these marginals are the same for all settings, as (1) requires, so the table is causal. The experiment being arbitrary, it follows that S is extensibly causal. □

To apply the lemma to our S we need one more ingredient, a decomposition of S that is almost, but not quite, of the form (3). Write

$$\begin{aligned} U = \tfrac{1}{20}(IIII + IZZI), \qquad V = \tfrac{1}{20}(IIII + ZIXZ), \\ C = \tfrac{1}{160}\,\mathrm{diag}(37,\, 7,\, 7,\, -3) \ \text{ on } A_I \otimes B_I, \end{aligned} \tag{16}$$

and $Q_C = C \otimes I_{A_O} \otimes I_{B_O}$. Since the first four terms of (9) are $\theta \otimes \theta$ on the inputs and $\theta \otimes \theta = 2C + \frac{1}{10} I \otimes I$,

$$S = U + V + 2Q_C. \tag{17}$$

Here $U \ge 0$ has order $A \prec B$ and $V \ge 0$ has order $B \prec A$, so (17) would be a causal decomposition if C were positive. It is not, but its negativity is mild: both marginals of C equal $\frac{1}{160}\,\mathrm{diag}(44, 4) \ge 0$, and this, together with the identity (23) below, is all we use.

Lemma 2. *Let S be the process* (9)*. For every ancilla state σ on $R_A \otimes R_B$ and every pair of instruments, with Choi matrices $M_{a|x}$ on $A_I \otimes R_A \otimes A_O$ and $N_{b|y}$ on $B_I \otimes R_B \otimes B_O$, the numbers*

$$\begin{aligned} r_A(a|x) &= \mathrm{Tr}\big[(\tfrac{1}{2}\theta \otimes \sigma_{R_A})\, E_{a|x}\big], \\ r_B(b|y) &= \mathrm{Tr}\big[(\tfrac{1}{2}\theta \otimes \sigma_{R_B})\, H_{b|y}\big], \end{aligned} \tag{18}$$

with $E_{a|x} = \mathrm{Tr}_{A_O} M_{a|x}$, $H_{b|y} = \mathrm{Tr}_{B_O} N_{b|y}$, $\sigma_{R_A} = \mathrm{Tr}_{R_B}\sigma$ and $\sigma_{R_B} = \mathrm{Tr}_{R_A}\sigma$, satisfy the hypotheses of Lemma 1 with $\lambda = \frac{1}{2}$.

Proof. The operators $E_{a|x}$ on $A_I \otimes R_A$ and $H_{b|y}$ on $B_I \otimes R_B$ are the measurement operators that give the outcome probabilities when the outputs are ignored; they are positive and sum to the identity over outcomes. Hence $r_A \ge 0$ depends only on x, $r_B \ge 0$ only on y, and each sums to $1/2$. It remains to prove (13). Fix α, β, let $M = \sum_{a\in\alpha} M_{a|x}$ and $N = \sum_{b\in\beta} N_{b|y}$, with $E = \mathrm{Tr}_{A_O} M$ on $A_I \otimes R_A$ and $H = \mathrm{Tr}_{B_O} N$ on $B_I \otimes R_B$, so that $0 \le E, H \le I$. Three facts are needed. First, the Born rule (2) for the extended process $S \otimes \sigma$, with (17) inserted, splits into three terms,

$$p(\alpha, \beta|x, y) = p_U + p_V + 2\,\mathrm{Tr}\big[(C \otimes \sigma)(E \otimes H)\big], \tag{19}$$

where $p_U = \mathrm{Tr}[(U \otimes \sigma)(M \otimes N)]$, p_V is defined likewise, and in the Q_C term the identity on the outputs has traced M and N down to E and H, so that this term is a trace over $A_I \otimes B_I \otimes R_A \otimes R_B$. Second, since U has order $A \prec B$, the joint probability p_U is at most the probability of Alice's event alone, and similarly for V:

$$p_U \le \mathrm{Tr}\big[(\tfrac{1}{10} I \otimes \sigma_{R_A})E\big], \quad p_V \le \mathrm{Tr}\big[(\tfrac{1}{10} I \otimes \sigma_{R_B})H\big]. \tag{20}$$

Here $\frac{1}{10} I$ is Alice's input state in U: replacing N by the full channel $\sum_b N_{b|y} \ge N$ can only increase p_U, and then $\mathrm{Tr}_{B_O} \sum_b N_{b|y} = I$ and $\mathrm{Tr}_{B_I} \frac{1}{20}(III + IZZ) = \frac{1}{10} II$ give the

stated value. Third, the reference marginals were built to contain exactly these bounds: since $\frac{1}{2}\theta = \frac{1}{10}I + \mathrm{Tr}_{B_I} C$, definition (18) gives

$$\begin{aligned} r_A(\alpha|x) &= \mathrm{Tr}\big[(\tfrac{1}{10}I \otimes \sigma_{R_A})E\big] + \mathrm{Tr}\big[(C \otimes \sigma)(E \otimes 1)\big], \\ r_B(\beta|y) &= \mathrm{Tr}\big[(\tfrac{1}{10}I \otimes \sigma_{R_B})H\big] + \mathrm{Tr}\big[(C \otimes \sigma)(1 \otimes H)\big]. \end{aligned} \tag{21}$$

Combining these equations gives the inequality

$$\begin{aligned} &r_A(\alpha|x) + r_B(\beta|y) - p(\alpha,\beta|x,y) \\ &\quad \geq \mathrm{Tr}\big[(C \otimes \sigma)(E \otimes 1 + 1 \otimes H - 2E \otimes H)\big]. \end{aligned} \tag{22}$$

The ordered parts U, V of S have dropped out entirely; only the signed part C remains to be controlled. As C is diagonal in the computational basis $|ij\rangle$ of $A_I \otimes B_I$, only the blocks $E_i = \langle i|E|i\rangle$, operators on R_A, and $H_j = \langle j|H|j\rangle$, operators on R_B, enter, and the right side is $\frac{1}{160}\mathrm{Tr}[\sigma\,(37D_{00} + 7D_{01} + 7D_{10} - 3D_{11})]$, a trace over $R_A \otimes R_B$, with $D_{ij} = E_i \otimes 1 + 1 \otimes H_j - 2E_i \otimes H_j$. The E_i, H_j are arbitrary operators with $0 \leq E_i, H_j \leq 1$ on the ancillas R_A, R_B, which have arbitrary dimension, and E_0, E_1 need not commute. Nevertheless, with $L = 7E_0 + E_1 - 7H_0 - H_1$ and $M = -E_0 + E_1 - H_0 + H_1$,

$$\begin{aligned} &37D_{00} + 7D_{01} + 7D_{10} - 3D_{11} \\ &\quad = 16(E_0 - H_0)^2 + \tfrac{1}{2}L^2 + \tfrac{7}{2}M^2 \\ &\qquad + 44\big[(E_0 - E_0^2) + (H_0 - H_0^2)\big] \\ &\qquad + 4\big[(E_1 - E_1^2) + (H_1 - H_1^2)\big], \end{aligned} \tag{23}$$

an identity between operators on $R_A \otimes R_B$ verified by expansion, the cross terms $E_0E_1 + E_1E_0$ and $H_0H_1 + H_1H_0$ cancelling between L^2 and M^2. Each square is positive and $E - E^2 \geq 0$ whenever $0 \leq E \leq 1$, so the right side of (22) is nonnegative and (13) holds. □

Lemmas 1 and 2 together show that S is extensibly causal, which completes the proof of the theorem. Nothing here is special to qubits: for any dimensions of the four spaces, all at least two, there is an open set of extensibly causal nonpurifiable processes, while if any space is one-dimensional every valid process is ordered and purifiable. The proof, which reuses Lemmas 1 and 2, is given in Appendix A.

DISCUSSION

We have exhibited a bipartite process whose every experiment, with or without shared entanglement, admits a causal explanation, and which nevertheless has no purification. This answers Problem 43 in the negative, in its finite-dimensional bipartite form and with the notion of purity of [4], and confirms the conjecture of [8] that causally nonseparable bipartite processes exist which remain causal under all entangled extensions. The lesson, we think, is that extensible causality and purifiability test different things: the former lets the causal explanation depend on the experiment, as (18) and (14) do through σ and the instruments, whereas the latter demands one explanation for all experiments, a decomposition of S itself. We do not claim that S is unphysical in every sense, only that the purification postulate is strictly stronger than the absence of causal-inequality violations.

Appendix A: General port dimensions, robustness, and the dimension boundary

Write $(m, d, n, e) = (\dim A_I, \dim A_O, \dim B_I, \dim B_O)$. For the robustness statement we need the linear constraints behind normalization explicitly. A Hermitian T on $A_I \otimes A_O \otimes B_I \otimes B_O$ satisfies $\mathrm{Tr}[T(M \otimes N)] = \mathrm{Tr}\,T/(de)$ for all pairs of channels if and only if [1, 10]

$$\begin{gathered} (1 - \mathcal{R}_{A_O})\mathcal{R}_{B_IB_O}T = 0, \qquad (1 - \mathcal{R}_{B_O})\mathcal{R}_{A_IA_O}T = 0, \\ (1 - \mathcal{R}_{A_O})(1 - \mathcal{R}_{B_O})T = 0, \end{gathered} \tag{A1}$$

with the replacement maps $\mathcal{R}_Y$ of the main text. Call the real space of such T $\mathcal{L}_V$; a valid process is $T \in \mathcal{L}_V$ with $T \geq 0$ and $\mathrm{Tr}\,T = de$. Replacement maps on different factors commute, preserve $\mathcal{L}_V$ and the trace, and are positive and unital, hence contractive in operator norm on Hermitian operators.

Now let $m, d, n, e \geq 2$. In each space single out two basis vectors $|0\rangle, |1\rangle$ and define, for $r \in \{m, d, n, e\}$,

$$\begin{aligned} P_r &= |0\rangle\langle 0| + |1\rangle\langle 1|, \qquad Z_r = |0\rangle\langle 0| - |1\rangle\langle 1|, \\ X_r &= |0\rangle\langle 1| + |1\rangle\langle 0|, \qquad Z_r^+ = I_r - 2|1\rangle\langle 1|, \\ \theta_r &= \tfrac{1}{4}\big(3|0\rangle\langle 0| + |1\rangle\langle 1|\big), \end{aligned} \tag{A2}$$

so that Z_r lives on the two distinguished vectors while Z_r^+ is an involution on the whole space. Put $\Pi = P_m \otimes I_d \otimes P_n \otimes I_e$, $G_A = P_m \otimes Z_d^+ \otimes Z_n \otimes I_e$, $G_B = Z_m \otimes I_d \otimes X_n \otimes Z_e^+$, and

$$\begin{aligned} \bar{S} &= \theta_m \otimes I_d \otimes \theta_n \otimes I_e + \tfrac{1}{20}(G_A + G_B), \\ \bar{S}_{\mathrm{c}} &= \tfrac{99}{100}\bar{S} + \tfrac{1}{100}\,\frac{I}{mn}. \end{aligned} \tag{A3}$$

For four qubits $\bar{S}$ is the process (9).

Theorem A. *(a) For all $m, d, n, e \geq 2$, $\bar{S}$ is a valid process of rank $4de$ that is extensibly causal and not causally separable. (b) Every $\bar{S}_{\mathrm{c}} + H$ with $H \in \mathcal{L}_V$, $\mathrm{Tr}\,H = 0$ and $\|H\|_\infty \leq 1/(300mn)$ is a full-rank extensibly causal process that is not causally separable. (c) If $\min\{m, d, n, e\} = 1$, every valid process has a definite causal order and is purifiable.*

By the theorem of Refs. [5, 6], "not causally separable" may be read as "not purifiable" in (a) and (b). Hence, on a given port tuple, every extensibly causal process is purifiable if and only if $\min\{m, d, n, e\} = 1$.

Proof of (a). Validity is checked as in the main text: with $\mathrm{Tr}_{A_O} M = I$ and $\mathrm{Tr}_{B_O} N = I$, the term G_A contributes $\mathrm{Tr}[(P_m \otimes Z_d^+)M]\ \mathrm{Tr}\,Z_n = 0$, the term G_B contributes zero by $\mathrm{Tr}\,Z_m = 0$, and the first term contributes $\mathrm{Tr}\,\theta_m\,\mathrm{Tr}\,\theta_n = 1$. The operator $\bar{S}$ vanishes outside the range of Π, which has dimension $4de$, and on it, in a basis of eigenvectors of Z_d^+ and

Z_e^+, it is a direct sum of the 2×2 blocks (10) acting on the two distinguished vectors of B_I; so $\bar S \geq 0$ with rank $4de$.

Lemma 1 is dimension-free. Lemma 2 and its proof go through verbatim with $U = \frac{1}{20}(\Pi + G_A)$, $V = \frac{1}{20}(\Pi + G_B)$, the same C, now a diagonal operator on $A_I \otimes B_I$ supported on the four vectors $|ij\rangle$ with $i, j \in \{0, 1\}$, and θ_m, θ_n in place of θ in (18). Indeed, the only facts about the qubit case that the proof used were $S = U + V + 2Q_C$, which holds for $\bar S$ because $\theta_m \otimes \theta_n = 2C + \frac{1}{10}P_m \otimes P_n$; that $U, V \geq 0$ have orders $A \prec B$ and $B \prec A$, which holds because $G_A^2 = G_B^2 = \Pi$ with both supported on the range of Π; the marginal $\mathrm{Tr}_{B_I} \frac{1}{20}(P_m \otimes I_d \otimes P_n + P_m \otimes Z_d^+ \otimes Z_n) = \frac{1}{10}P_m \otimes I_d$, which replaces $\frac{1}{10} I \otimes I$ in (20); the identity $\frac{1}{2}\theta_m = \frac{1}{10}P_m + \mathrm{Tr}_{B_I} C$; and the fact that only the blocks $\langle i|E|i\rangle$, $\langle j|H|j\rangle$ with $i, j \in \{0, 1\}$ enter the C term, so that the identity (23) applies unchanged. Hence $\bar S$ is extensibly causal.

For nonseparability let $P_1 = |1\rangle\langle 1|$ on A_I and $D_n = I_n + 8|1\rangle\langle 1|$ on B_I, and take

$$\begin{aligned}\bar W = {} & P_1 \otimes P_d \otimes D_n \otimes P_e - P_1 \otimes Z_d \otimes D_n Z_n^+ \otimes P_e \\ & + 3\, P_1 \otimes P_d \otimes X_n \otimes Z_e,\end{aligned} \tag{A4}$$

which reduces to (11) for qubits. Applying $\mathcal{R}_{B_O}$ kills the last term, and on the eigenspaces of Z_d the operator on B_I is $D_n(I_n \mp Z_n^+)$, equal to $18|1\rangle\langle 1|$ or $2(I_n - |1\rangle\langle 1|)$, both positive. Applying $\mathcal{R}_{A_O}$ kills the middle term, and on the eigenspaces of Z_e the operator on B_I is $D_n \pm 3X_n$, whose restriction to the distinguished vectors is the rank-one positive matrix of the main text and whose remainder is the identity. So $\bar W$ is a causal witness, and the three terms of (A4) pair with the three parts of (A3) to give $\mathrm{Tr}(\bar W \bar S) = 3 - 2 - \frac{6}{5} = -\frac{1}{5}$, exactly as in (12). □

Proof of (b). For Hermitian $X \in \mathcal{L}_V$ with $\mathrm{Tr}\, X = 0$ set

$$\begin{aligned} X_A &= \mathcal{R}_{B_O} X - \tfrac{1}{2}\mathcal{R}_{A_O B_O} X, \\ X_B &= \mathcal{R}_{A_O} X - \tfrac{1}{2}\mathcal{R}_{A_O B_O} X. \end{aligned} \tag{A5}$$

The last equation of (A1) gives $X_A + X_B = X$; the properties of replacement maps listed above give $X_A, X_B \in \mathcal{L}_V$, traceless, with $\mathcal{R}_{B_O} X_A = X_A$ and $\mathcal{R}_{A_O} X_B = X_B$, and $\|X_A\|_\infty, \|X_B\|_\infty \leq \frac{3}{2}\|X\|_\infty$. Hence if $\|X\|_\infty \leq 1/(3mn)$, then $I/(mn) + X = T_A + T_B$ with $T_A = I/(2mn) + X_A \geq 0$ of order $A \prec B$, $T_B = I/(2mn) + X_B \geq 0$ of order $B \prec A$, and $\mathrm{Tr}\, T_A = \mathrm{Tr}\, T_B = de/2$. Apply this to $X = 100H$:

$$\bar S_{\mathrm{c}} + H = \tfrac{99}{100}\bar S + \tfrac{1}{100}(T_A + T_B). \tag{A6}$$

In any experiment the table of (A6) is $\frac{99}{100}p_{\bar S} + \frac{1}{100}(p_{T_A} + p_{T_B})$. By Lemma 1, $p_{\bar S} = \frac{1}{2}p^{A\prec B} + \frac{1}{2}p^{B\prec A}$, while $2p_{T_A}$ and $2p_{T_B}$ are normalized tables of orders $A \prec B$ and $B \prec A$ (an ordered process stays ordered under $\otimes\sigma$). Grouping by order, the table is a mixture with weight $\frac{1}{2}$ of two one-way tables whose marginals depend on one setting only, so $\bar S_{\mathrm{c}} + H$ is extensibly causal. It is full rank, since $\bar S \geq 0$ and $\|H\|_\infty \leq 1/(300mn)$ give $\bar S_{\mathrm{c}} + H \geq I/(150mn)$. Finally, the spectrum of $\bar W$ consists of $9 \pm 3\sqrt{10}$ and $1 \pm \sqrt{10}$, each twice, and 2 with multiplicity $2(n-2)$, so $\mathrm{Tr}\, \bar W = 4(n+8)$ and $\|\bar W\|_1 = 16\sqrt{10} + 4(n-2)$; for $m, n \geq 2$,

$$\begin{aligned} \mathrm{Tr}(\bar W \bar S_{\mathrm{c}}) &= -\tfrac{99}{500} + \tfrac{n+8}{25mn} \leq -\tfrac{49}{500}, \\ |\,\mathrm{Tr}(\bar W H)| &\leq \frac{\|\bar W\|_1}{300mn} \leq \frac{\sqrt{10}}{75}, \end{aligned} \tag{A7}$$

whence $\mathrm{Tr}[\bar W(\bar S_{\mathrm{c}} + H)] \leq -(147 - 20\sqrt{10})/1500 < 0$. □

Proof of (c). If $d = 1$, then $\mathcal{R}_{A_O}$ is the identity map, so every $T \in \mathcal{L}_V$ satisfies $\mathcal{R}_{A_O} T = T$ and has order $B \prec A$; $e = 1$ gives order $A \prec B$ likewise. If $m = 1$, the second equation of (A1) reads $(1 - \mathcal{R}_{B_O})\mathcal{R}_{A_O} T = 0$, and adding the third gives $(1 - \mathcal{R}_{B_O})T = 0$: order $A \prec B$. The case $n = 1$ is symmetric.

It remains to purify an ordered process, say $S = \widehat{S} \otimes I_{B_O}$ with $\mathrm{Tr}_{B_I} \widehat{S} = \rho \otimes I_{A_O}$ as in the main text. Let $\rho = \sum_{i=1}^r \lambda_i |i\rangle\langle i|$ with all $\lambda_i > 0$, let R be a copy of the support of ρ, and let $|\psi\rangle = \sum_{i=1}^r \sqrt{\lambda_i}\, |i\rangle_{A_I} |i\rangle_R$. Positivity of $\widehat{S}$ and the marginal condition force $\widehat{S}$ to vanish on $\ker\rho \otimes A_O \otimes B_I$, so $J = (\rho^{-1/2} \otimes I)\widehat{S}(\rho^{-1/2} \otimes I)$ is well defined on $R \otimes A_O \otimes B_I$, positive, with $\mathrm{Tr}_{B_I} J = I_R \otimes I_{A_O}$: it is the Choi matrix of a channel $C\colon R \otimes A_O \to B_I$, and preparing $|\psi\rangle$ and then applying C to $R \otimes A_O$ reproduces $\widehat{S}$. Dilate C to an isometry $V\colon R \otimes A_O \to B_I \otimes E_1$ and complete it to a unitary $U_1\colon R \otimes A_O \otimes E_0 \to B_I \otimes E_1$ with $U_1(\xi \otimes |0\rangle) = V\xi$, padding E_0 and E_1 so that the dimensions match. Take a past P with a unitary $U_0\colon P \to A_I \otimes R \otimes E_0$ sending $|0\rangle$ to $|\psi\rangle|0\rangle$, and a future $F = B_O \otimes E_1$. The circuit "U_0; Alice on A_I; U_1 on $R \otimes A_O \otimes E_0$; Bob on B_I; output $B_O \otimes E_1$" is a composition of unitaries whenever the local operations are unitary, also when they act on external systems, so it is a pure process; feeding $|0\rangle_P$ and discarding F returns S. The opposite order is obtained by exchanging the parties. □

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